



Effects of *Ferula asafetida*, closantel, albendazole, oxfendazole, and ivermectin against *Haemonchus contortus* in goats and sheep

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Abstract

Background Haemonchosis is a fatal disease of small ruminants caused by the parasite *Haemonchus contortus* (*H. contortus*). The most common drugs used in the treatment of *H. contortus* include albendazole, oxfendazole, and ivermectin. However, as previously reported in the treatment of haemonchosis, these medicines have acquired drug resistance problems over time. Interestingly, natural plant compounds have demonstrated promising effects in the treatment of *H. contortus*. Therefore, the current study evaluated the effects of plant extract, *Ferula asafetida*, against common drugs such as albendazole, oxfendazole, ivermectin, and closantel for the treatment of haemonchosis in small ruminants.

Methodology The current study was conducted on different small ruminant farms in Kasur District, Punjab, Pakistan. The positive animals ($n = 720$) after coprological examination were selected in this study and divided into two major groups ($n = 360$ goats and $n = 360$ sheep). Further, animals were divided into five treatment groups (A–E) and one control group with no treatment (F). Albendazole, oxfendazole, ivermectin, closantel, and *Ferula asafetida* were administered orally to groups A–E, respectively. The eggs per gram feces (EPG) were determined through the McMaster technique on days 0, 7th, and 14th of treatment.

Results The results showed a significantly higher efficacy of closantel and *Ferula asafetida* against *H. contortus* in both goats (100% and 70%; $p < 0.05$) and sheep (99% and 87%; $p < 0.05$), respectively. No correlation was observed between EPG reduction with age and gender in both goats and sheep.

Conclusion Allopathic drug closantel and herbal drug, *Ferula asafetida*, have been proved an effective dewormer against *H. contortus* in small ruminants.

Keywords Haemonchosis · Small ruminants · Resistance · Closantel · *Ferula asafetida*

Introduction

Small ruminants (sheep and goats) are the major livestock species raised for many purposes, i.e., milk, meat, and hides. Pakistan is the 3rd largest goat-populated country globally, having 78 million heads (Jabbar et al. 2020; Shah et al. 2017). However, the production of small ruminants is decreasing day by day worldwide due to infectious and non-infectious diseases (Rizwan et al. 2017). Gastrointestinal parasitic infection in small ruminants is one of the significant health issues in Pakistan and worldwide (Asif

et al., 2008). Haemonchosis in the small ruminants is caused by blood-sucking gastrointestinal roundworm *Haemonchus contortus* (*H. contortus*) (Dutta et al., 2017). Adult parasite sucks blood (0.05 ml blood/day/parasite). It damages the host's intestinal walls (Hu and Alam 2019); severe infestations of this helminths lead to anemia, hypoproteinemia, edema, pregnancy toxemia, diarrhea, severe emaciation, and high mortality in sheep and goats (Besier et al. 2016; Dey et al. 2019; Dutta et al. 2017). The high prevalence of this parasite in small ruminants has been reported worldwide and in Pakistan (Al-Shaibani et al. 2008, Asif et al. 2008, Dey, Zhang, Begum, Alim, Hu and Alam 2019, Qasim et al. 2016). Several broad-spectrum anthelmintics are used in Pakistan and worldwide (Iqbal et al., 2017; Kotze and Prichard, 2016). Failure of anthelmintic and low efficacy

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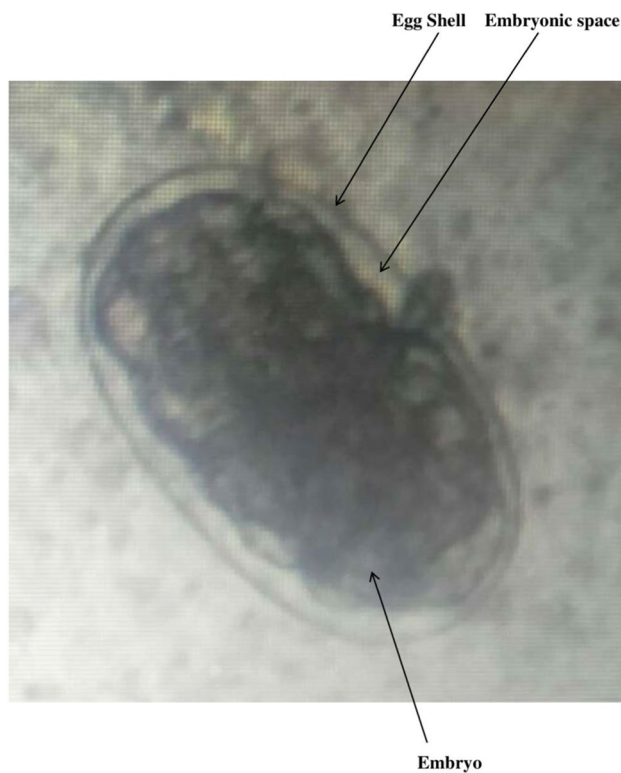


Fig. 1 *Haemonchus contortus* egg under the microscope

of drugs are mainly due to anthelmintic resistance in goats and sheep (Shalaby, 2013).

Anthelmintic resistance was reported in the nineties, creating great problems in the small ruminants (Das et al., 2015). Among many species of gastrointestinal nematodes, the resistance of variable degree has been developed in the warm and humid parts of the world due to the frequent and low dose of the same drug (Fleming et al. 2006). Usually, resistance appears within 10 years of introducing a new drug (Kotze and Prichard, 2016). Recently, many studies have been done on anthelmintic resistance in many parts of

the world, mainly in Africa (Howell et al., 2008; Basripuzi et al., 2012). Anthelmintics belonging to benzimidazole and tetrahydropyrimidine (ivermectin) group showed high efficacy against nematodes in goats' in vivo and in vitro trials (Adediran and Uwalaka 2015; Das et al., 2015). However, these drugs are not used appropriately. Therefore, resistance have been developed against them ((Adediran and Uwalaka 2015; Das et al. 2015; Gonzalez et al. 2019; Kaplan and Vidyashankar 2012; Wang et al. 2017). The population of *H. contortus* species shows resistance to major anthelmintic classes like benzimidazole, macrocyclic lactones, and imidazothiazoles (Iqbal et al. 2017; Kotze and Prichard 2016; Shalaby 2013). Resistance is the main issue with allopathic drugs, whereas this is absent in herbal drugs (Hamad 2014).

Many natural plant products in herbal drugs are effective against gastrointestinal nematodes (Hamad 2014). *Ferula asafetida* (Hing) is a herbal drug; it has traditionally been used for its anthelmintic properties in many countries such as Iran, China, and Nepal, where it has been used to treat intestinal infection parasites (Bhatnager et al. 2015; Şahin et al. 2003). Therefore, we hypothesized that *Ferula asafetida* would be very effective against *H. contortus* with high efficacy than any allopathic (common dewormer) in small ruminants. This study aims to determine the efficacy of common drugs (albendazole, oxfendazole, ivermectin, and closantel) and *Ferula asafetida* against *H. contortus* and to establish the standard protocol for treatment and control of *H. contortus* in goats and sheep.

Material and method

Experimental sites and animals

The current study was conducted in goats and sheep farms in Kasur District, Pakistan. A total of 720 *H. contortus* positive sheep ($n=360$) and goats ($n=360$) were selected. The selected

Table 1 Efficacy of common dewormer and *Ferula asafetida* against *H. contortus* in goats and sheep

Treatment groups	Sheep (%)	Goats (%)
Group A	9	39
Group B	4	35
Group C	17	22
Group D	99**	100**
Group E	87**	70*
Control	0	0

Comparative overall efficacy of anthelmintics in sheep and goats, groups A to F show albendazole, oxfendazole, ivermectin, closantel, *Ferula asafetida*, and control groups, respectively. ** denoted $p < 0.001$ and * denoted $p < 0.01$

Table 2 Anthelmintic efficacy against *H. contortus* in sheep and goats with relationship to gender

Treatment groups	Goats (%)		Sheep (%)	
	Male	Female	Male	Female
Group A	40	39	9	9
Group B	33	33	4	4
Group C	22	22	17	17
Group D	100	100	99	99
Group E	70	70	87	87
Group F	0	0	0	0

Groups A to F show albendazole, oxfendazole, ivermectin, closantel, *Ferula asafetida*, and control groups

Table 3 Comparative anthelmintic efficacy against *H. contortus* in goats and sheep among different age groups

Treatment groups	Goats (%)			Sheep (%)		
	0–2 yrs	2–3 yrs	3–4 yrs	0–2 yrs	2–3 yrs	3–4 yrs
Group A	39	39	39	8.5	9	9
Group B	33	32	33	4.3	4	4
Group C	23	22	22	18	17	17
Group D	100	100	100	99	99	99
Group E	70	70	70	87	87	87
Group F	0	0	0	0	0	0

Groups A to F show albendazole, oxfendazole, ivermectin, closantel, *Ferula asafetida*, and control groups. Anthelmintic efficacy against *H. contortus* is shown in percentage (%)

Table 4 Effect of anthelmintic on reduction of EPG on various days in goats

Treatment groups	Day 0	Day 7	Day 14
Group A	1215.12 ± 344.37 ^Z	969.88 ± 278.81 ^{bcY}	730.88 ± 203.06 ^{bcX}
Group B	1462.12 ± 481.06 ^Z	1164.63 ± 384.18 ^{cdY}	937.13 ± 322.68 ^{cdX}
Group C	1409.87 ± 456.21 ^Z	1257.00 ± 409.24 ^{cdY}	1096.50 ± 356.73 ^{cdX}
Group D	1508.75 ± 407.11 ^Z	104.13 ± 28.11 ^{aY}	0.00 ^{aX}
Group E	1468.00 ± 342.53 ^Z	498.50 ± 116.69 ^{abY}	439.13 ± 103.63 ^{bX}
Control	1466.75 ± 340.27	1465.63 ± 337.85 ^d	1537.13 ± 365.13 ^d

Groups A to F show albendazole, oxfendazole, ivermectin, closantel, *Ferula asafetida*, and control groups. a, b, c, and d show the level of significance between treatment groups, while X, Y, and Z show level of significance difference within treatment groups

Table 5 Effect of anthelmintic on reduction of EPG on various days in sheep

Treatment groups	Day 0	Day 7	Day 14
Group A	1747.13 ± 679.41 ^Y	1663.62 ± 646.91 ^{bY}	1582.25 ± 632.73 ^{bY}
Group B	1816 ± 729.61 ^Y	1781 ± 716.18 ^{bY}	1734.75 ± 699.73 ^{bY}
Group C	1738.25 ± 632.46 ^Y	1573.81 ± 578.02 ^{bY}	1432.75 ± 466.61 ^{bY}
Group D	1873.51 ± 638.91 ^Y	147.13 ± 51.13 ^{aX}	17.75 ± 5.65 ^{aX}
Group E	1873.63 ± 607.19 ^Y	399.12 ± 143.97 ^{aX}	243.54 ± 78.91 ^{aX}
Control	1884.88 ± 622.89	1910 ± 607.49 ^b	1951.51 ± 630.06 ^b

Groups A to F show albendazole, oxfendazole, ivermectin, closantel, *Ferula asafetida*, and control groups. a and b show the level of significance between treatment groups, while X and Y show level of significance difference within treatment groups

animals were subdivided based on gender (male and female) and age (0–2, 2–3, and 3–4 years). The animals who had free access to fresh drinking water could feed on green fodder twice a day. The ethical committee of the University of Veterinary and Animal Sciences (UVAS), Lahore, Pakistan, approved the experimental design and all study procedures.

Experimental design and anthelmintic treatment

The randomly selected animals were divided into six groups (A–F). Each group consists of 30 goats and 30 sheep. The animals of group A were treated with albendazole bolus at a rate of 10 mg/kg. Group B was given oxfendazole bolus at a 5 mg/kg rate, while group C was given a bolus of ivermectin

at a dose rate of 0.02 mg/kg. Closantel bolus was given to the animals of group D at a dose rate of 10 mg/kg, while group E was given *Ferula asafetida* extract orally at the rate of 100 mg/kg. However, group F was not treated and considered as a control group. The animals of respective groups were treated at day 0 after sample collection. Fecal samples were collected at day 0 (pre-treatment), at day 7 and 14 (post-treatment) of all animals (Dixit et al. 2019; dos Santos et al. 2017; Puspitasari et al. 2016). *Ferula asafetida* whole plant was collected and dried under shade for several days. The dried plants were grounded with a hammer mill, then soaked in water for 7 days at room temperature. It was then filtered through a 2-mm sieve and collected the extract for further trial.

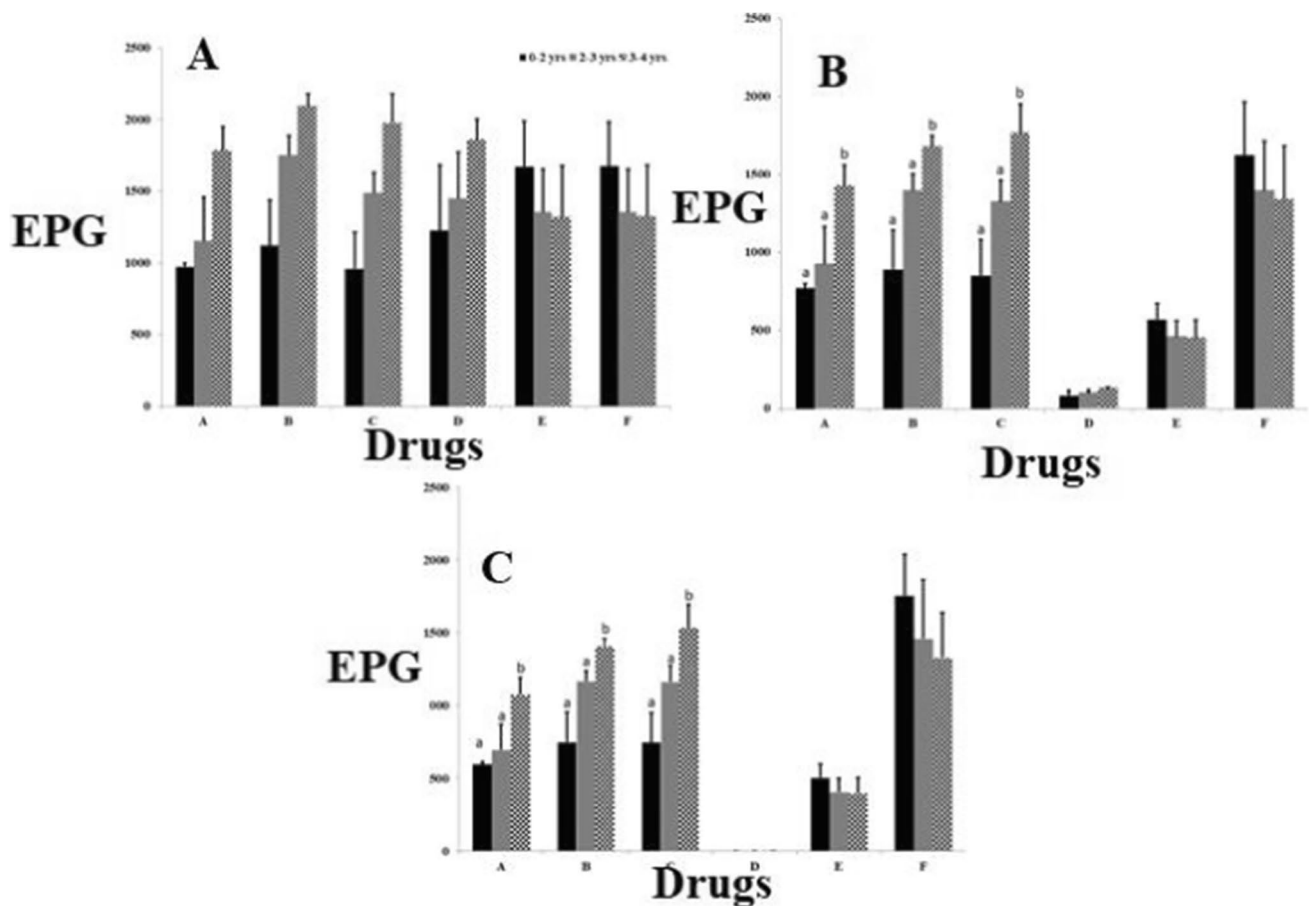


Fig. 2 Comparative reduction of EPG based on age-wise on various days in goats. **A** Day 0 of treatment, **B** day 7 after treatment, and **C** day 14. Groups A to F show albendazole, oxfendazole, ivermectin,

closantel, *Ferula asafetida*, and control groups. Different superscripts show the significance level in reducing EPG within the treatment group in different age-wise animals

Sample collection and fecal egg count

Fecal samples were collected directly from the rectum of animals in air-tight plastic tubes and transported to University Diagnostic Lab, University of Veterinary and Animal Sciences Ravi campus Pattoki. Samples were processed using parasitology techniques (sedimentation and flotation) and later identified as positive/negative samples. McMaster's quantitative examination was carried out to concentrate eggs as described elsewhere (Rizwan, Sajid, Iqbal and Saqib 2017). The parasite was known as described previously by identifying the ova or egg of *H. contortus* (Ljungström et al., 2018). The representative egg of *H. contortus* is shown in Fig. 1.

Statistical analysis

The data were statistically analyzed through statistical software SPSS version 21. The chi-square (χ^2) was used to test the efficacy of different dewormers, while repeated

measure ANOVA was used to test the EPG count gender and age-wise.

Result

In the present study, the effectiveness of albendazole, oxfendazole, ivermectin, closantel, and *Ferula asafetida* was evaluated against *H. contortus* based on the decrease of EPG in goats and sheep. The results showed that there was significantly higher ($p < 0.001$) efficacy of closantel (100%) as compared to another anthelmintic in goats (Table 1). The data of anthelmintic efficacy showed that in sheep, closantel and *Ferula asafetida* were 99% and 87%, respectively, which is highly effective compared to the other anthelmintics and control group.

Furthermore, the results showed that goats and sheep were not significantly different in anthelmintic efficacy and between males and females in all the treatment groups (Table 2). Similarly, there was no significant difference

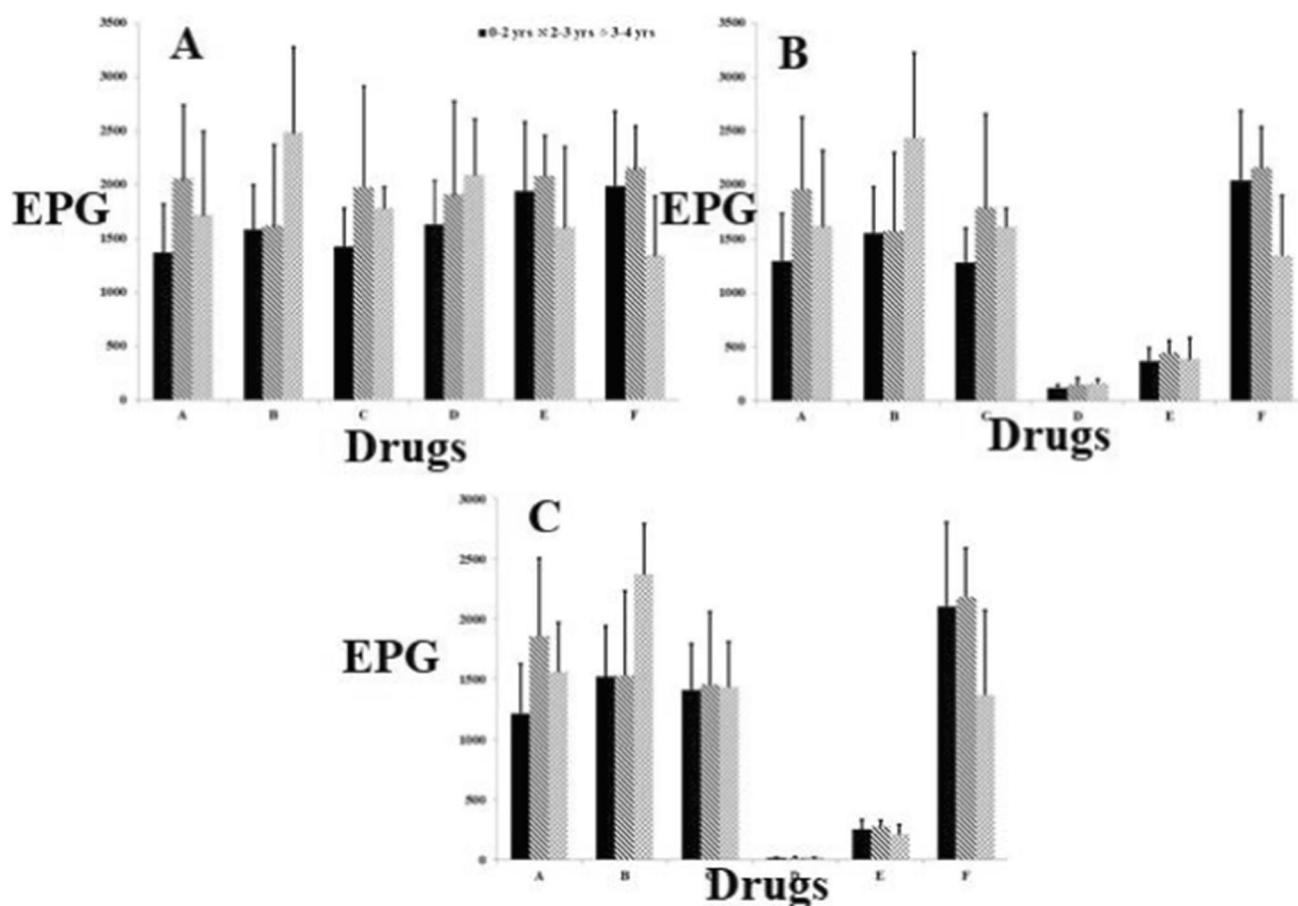


Fig. 3 Comparative reduction of EPG based on age-wise on various days in ovine. **A** Day 0 of treatment, **B** day 7 after treatment, and **C** day 14. Groups A to F show albendazole, oxfendazole, ivermectin, closantel, *Ferula asafetida*, and control groups

in the effectiveness of anthelmintic between different age groups in both goats and sheep, as shown in Table 3.

Effect of drugs on reduction of EPG on various days in goats and sheep

The reduction in the EPG in goats showed a significantly higher reduction of EPG in closantel treated at 7 and 14 days (104.13 ± 28.11 and 0) compared with other treated groups. All treated groups recorded the lowest decrease of EPG in the ivermectin group at 7 and 14 days (1257.00 ± 409.24 and 1096.50 ± 356.73), respectively. Moreover, in all treated groups, therapeutic data showed a significant reduction of EPG from day 0 to day 14 compared to control (F) and is presented in Table 4.

The result of the reduction in the EPG in sheep showed that there was a significantly higher reduction of EPG from day 0 to day 14 in closantel and *Ferula asafetida* (1873.51 ± 638.91 to 17.75 ± 5.65 and 1873.63 ± 607.19 to 243.54 ± 78.91) respectively in treated groups compared with other anthelmintic treatments. There was no significant

difference between albendazole, oxfendazole, and ivermectin treatment in EPG reduction at 7 and 14 days, as presented in Table 5.

However, there was no significant difference in EPG reduction between males and females of studied animals in all treatment groups (Fig. 2 and 3). Similarly, there was no significant difference in EPG reduction between the age of 0–2, 2–3, and 3–4 years of studied animals in all treatment groups, as shown in Figs. 4 and 5.

Discussion

Traditionally anthelmintics are used for the control of helminth in grazing animals and to reduce production losses. Regular use of un-selective and similar modes of action anthelmintics with improper dose rates for many years cause resistance issues (Coles and Roush 1992; Islam et al. 2018). Therefore in this study, four standard and one herbal anthelmintic were used to evaluate the efficacy and resistance in goats and sheep. To the best of our knowledge, this was the

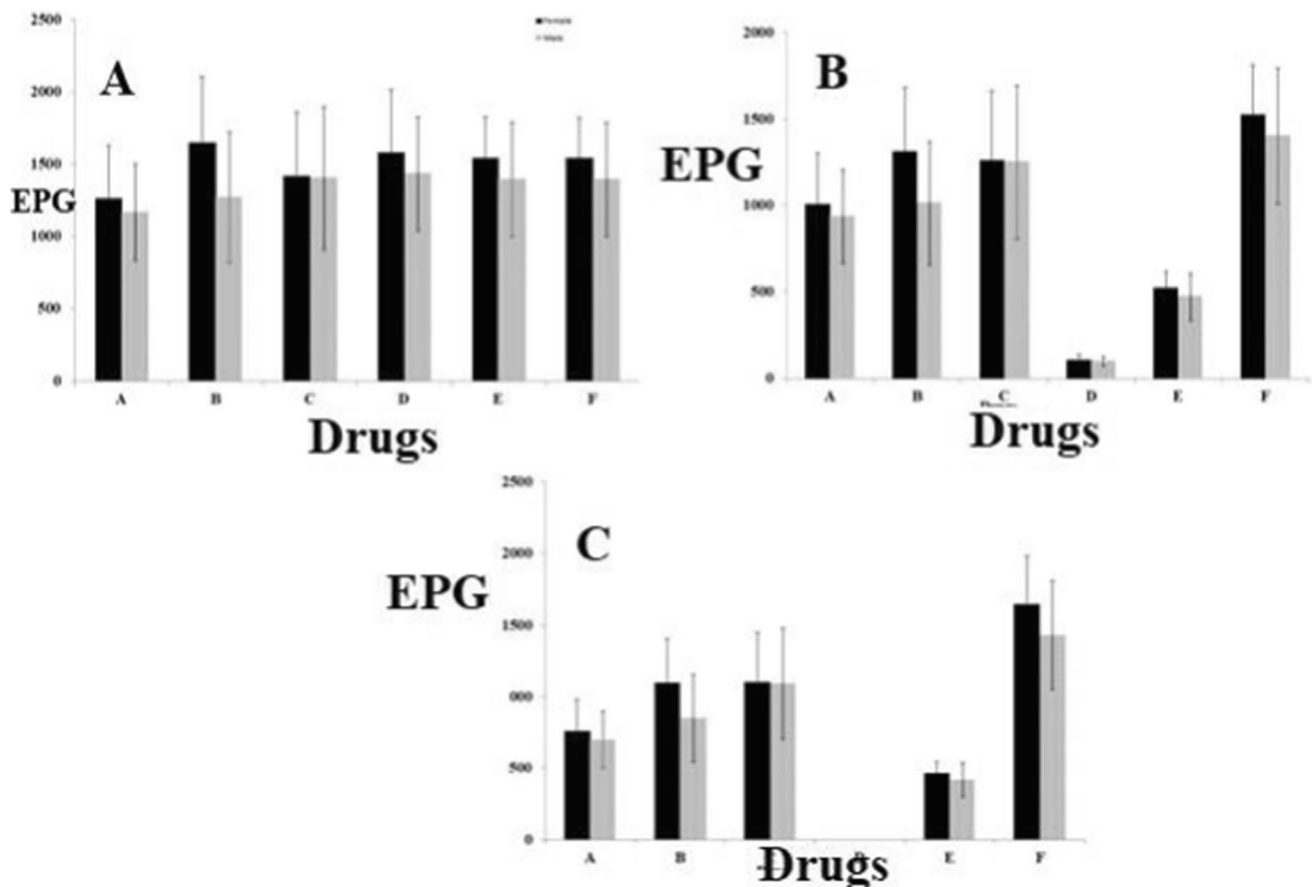


Fig. 4 Comparative reduction of EPG based on gender-wise on various days in goats. **A** Day 0 of treatment, **B** day 7 after treatment, and **C** day 14. Groups A to F show albendazole, oxfendazole, ivermectin, closantel, *Ferula asafetida*, and control groups

first study in Pakistan in which the efficacy of anthelmintics was determined in goats and sheep. The main finding of this study was that closantel and *Ferula asafetida* had higher drug efficacy irrespective of age and gender in studied animals. The EPG reduction was higher in closantel in both goats and sheep, while *Ferula asafetida* significantly reduced EPG in sheep than goats.

The current study results showed that the EPG reduction was highest in goats at the 7th and 14th days post-treatment in closantel than other anthelmintic drugs. Therefore the efficacy of closantel was 100% than the other anthelmintics against *H. contortus*. Our result was in agreement with the previous result, which showed that closantel had higher efficacy and low anthelmintic resistance in goats (Das, Dixit, Nath, Agrawal and Dongre 2015, Uppal et al. 1993). They found closantel to be 100% effective against *H. contortus* in goats using 10 mg/kg body weight. Dixit, Das, Dixit and Sharma (2019) found that closantel is 95.64% effective against *H. contortus*. The higher efficacy of closantel against *H. contortus* might be attributed to high binding properties to a blood protein important for oxidative phosphorylation in parasites that decreased the adenosine triphosphate

and nicotinamide in the mitochondria. Therefore, lower the availability of energy to parasites (Lanusse et al. 2009). However, some previous studies observed that helminths developed resistance against closantel (Baihaqi et al. 2019; Chandrawathani et al. 2013; Premaalatha et al. 2014). The current study showed that the efficacy of *Ferula asafetida* was 70% in goats and is lower than closantel and higher than other anthelmintics (Singh et al. 2016). It was also reported that 100% efficacy of aqueous extract of *Zanthoxylum armatum* DC seeds against *H. contortus* (Akhtar et al. 2000) observed that different medicinal plants had high efficacy against anthelmintics in goats of Indo-Pakistan. The high effectiveness of these medicinal plants might be the high contents of phenolic flavonoid and tannin that improved the host's immune system to destroy the parasites (Atoui et al. 2005; Tiwary et al. 2007). The other reason for low anthelmintic resistance is the high concentration of copper and zinc in these plants, which is high wormicidal and plays an indirect role in improving the host's immune system (Burke et al. 2005; Chartier et al. 2001; Waruiru et al. 2004). In this study, we found high anthelmintic resistance against albendazole, oxfendazole, and ivermectin, in agreement with

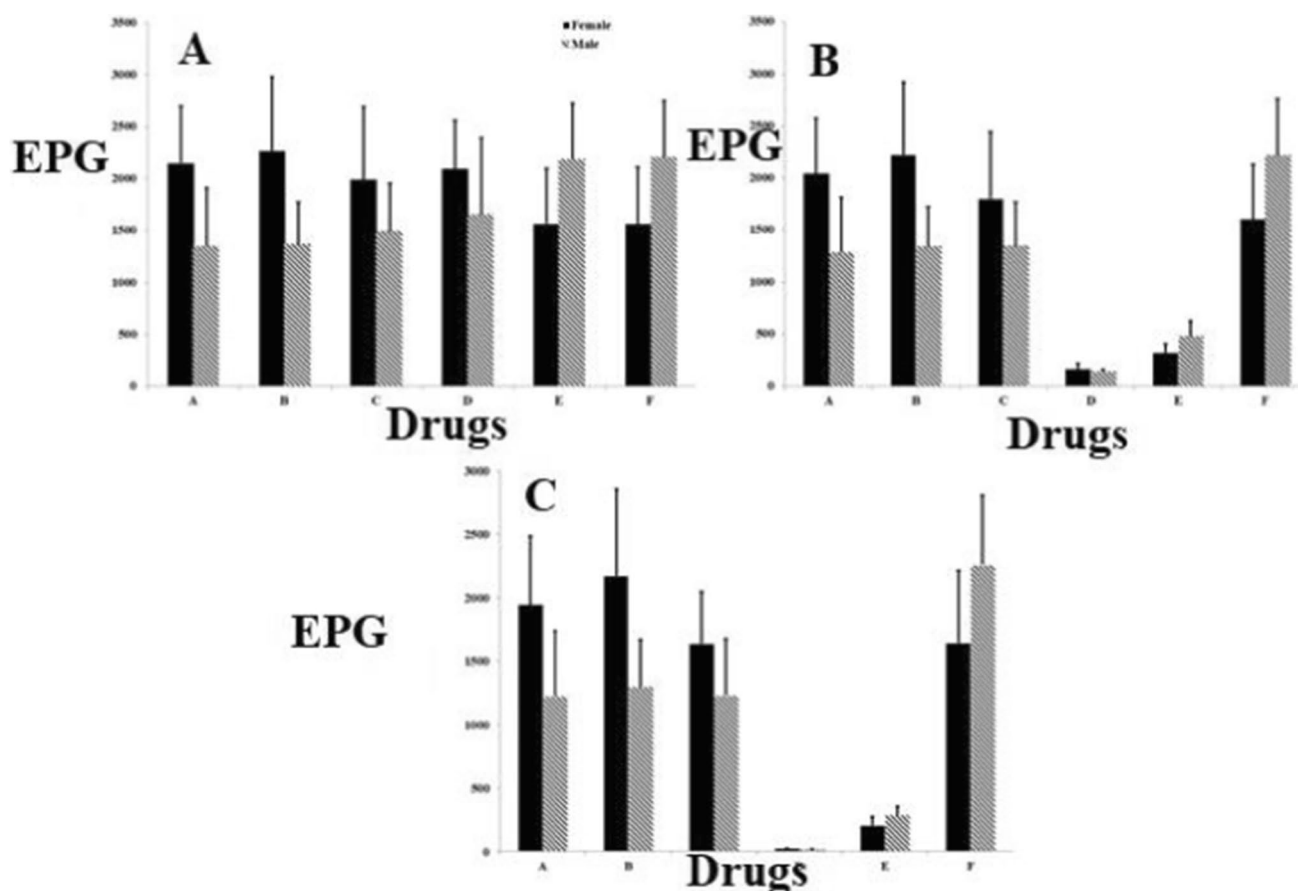


Fig. 5 Comparative reduction of EPG based on gender-wise on various days in sheep. **A** Day 0 of treatment, **B** day seven after treatment, and **C** day 14. Groups A to F show albendazole, oxfendazole, ivermectin, closantel, *Ferula asafetida*, and control groups

previous reports (Kumar Jaiswal et al. 2013; Maharshi et al. 2011; Uppal et al. 1993).

The current study's finding showed that closantel and *Ferula asafetida* had higher efficacy against *H. contortus* in sheep. Our result is in agreement with previous studies that reported closantel had high efficacy against *H. contortus* and other *Strongyloides* in native breeds of sheep (Westers et al. 2016) and merino sheep (Kadam et al. 2009; Shahana 2013; Trambo et al. 2017). They observed 100% and 98% efficacy of closantel against *H. contortus* at the 8th and 14th days of post-treatment in Indian sheep. The higher efficacy of closantel against *H. contortus* may be due to lowered population of *H. contortus* eggs and improvement of sheep's immune system. In literature, ivermectin had higher efficacy (99%) against *Strongyloides* and *H. contortus* in sheep when injected subcutaneously (Canga et al. 2009; Westers, Jones-Bitton, Menzies, Van Leeuwen, Poljak and Peregrine 2016). The reason for the higher efficacy of subcutaneous ivermectin might be the higher bioavailability of ivermectin to plasma. The recent literature review reports that aqueous extract of herbal plants had higher efficacy against *Strongyloides* and *H. contortus* (de Souza Chagas

and da Silva Vieira 2007; Dongre et al. 2015). The higher efficacy of herbal plants is due to the increased copper and zinc that is lethal for the growth and production of worm eggs (Chartier, Soubirac, Pors, Silvestre, Hubert, Couquet and Cabaret 2001, Waruiru, Mutunu and Otieno 2004). A similar finding has been reported by Danquah et al. (2012) that supplementation of copper and zinc had a significant reduction of *Strongyloides* eggs in the host. In the present study, albendazole, oxfendazole, and ivermectin had lower efficacy. Our finding was in agreement with previous reports that reported lower efficacy of albendazole and fenbendazole due to higher anthelmintic resistance in small ruminants (Baihaqi et al. 2019; Shahardar et al. 2014; Westers, Jones-Bitton, Menzies, Van Leeuwen, Poljak and Peregrine 2016). The higher anthelmintic resistance was due to the higher frequency of deworming and a low dose of anthelmintics (Sissay et al. 2006).

The current study found no significant difference in age and gender-wise efficacy of anthelmintics in both goats and sheep. This might be due to similar improvements of the immune defense system against *H. contortus* in the host. The reduction of eggs of *H. contortus* was significantly higher in closantel and

Ferula asafetida in both goats and sheep, regardless of age and gender. Our findings support the previous results that higher reduction of eggs in closantel and plants extract (leaves) (Dixit, Das, Dixit and Sharma 2019, Nabukenya et al. 2014, Trambo, Shahardar, Allaie, Wani and Abbas 2017). Albendazole, oxfendazole, and oral ivermectin had lowered the reduction of eggs in the current study. These might be due to the low availability of the drug in the plasma, which might have more resistant gene expression (Coles et al. 2006, Kenyon and Jackson 2012, Van den Brom et al. 2015). A longer-term study would be necessary to evaluate the resistance genes in the *H. contortus*.

Conclusion

In conclusion, the present study shows a higher reduction of *H. contortus* eggs after treatment with closantel in goats and a higher reduction of *H. contortus* eggs after treatment with closantel and *Ferula asafetida* in sheep. Therefore, higher efficacy of closantel against *H. contortus* is displayed in both goats and sheep. However, *Ferula asafetida* displayed only higher efficacy in sheep.

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Author contribution HIA conceived and designed the research. AJ and MZI conducted experiments. MA and MNW analyzed data. AJ, HS, MI, and MSZG wrote the manuscript. All authors read and approved the manuscript. AZD and AU revised the manuscript.

Data availability The numerical data used to support the findings of this study are available from the corresponding author upon request.

Declarations

Ethics approval The authors have completely observed ethical issues (including plagiarism, informed consent, misconduct, data fabrication and/or falsification, double publication and/or submission, redundancy, etc.).

Consent to participate Not applicable.

Consent for publication Not applicable.

Conflict of interest The authors declare no competing interests.

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